

Tien Nguyen

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Personal Profile

Systems software engineer with hands-on experience across the low-level stack and a strong grounding in operating-system internals. Comfortable in C, C++, Go, Rust, Python, and Matlab. More details on my previous works can be found [here](#).

Work Experience

Qualcomm Technologies

System Software Engineer

Cork, Ireland

May 2024 - Current

- Architect and ship production software for low-power, on-device machine learning solutions used in customer-facing applications on Qualcomm chipsets.
- Optimize ML models (IMU-only state estimation, human activity recognition, on-device personalization) for deployment on resource-constrained environments.
- Verify system functionality, troubleshooted technical issues, and provided ongoing customer support.

Programming Languages: C, C++, Python, Matlab.

Tools and Libraries: [Qualcomm AI Hub](#), [Qualcomm® AI Engine Direct](#), libTorch, LiteRT, ONNX, etc.

Vietnam—Korea Institute of Science and Technology

Robotics Software Engineer

Hanoi, Vietnam

Aug 2021 - May 2022

- Developed software, including actuator controller, safety management system, and graphical user interface, for a robotic platform.
- Collaborated in a cross-disciplinary environment.

Programming Languages: C, C++, Python.

Tools and Libraries: Robot Operating System (ROS), Qt.

Viettel High Technology Industries Corporation

Embedded Software Engineer

Hanoi, Vietnam

Aug 2020 - Aug 2021

- Developed and maintained Linux kernel device drivers for custom telecom hardware.
- Performed board bring-up: ported and customized the Das U-Boot bootloader, authored Device Tree descriptions, and configured, cross-compiled, and booted the Linux kernel on new hardware.

Programming Languages: C, C++.

Tools and Libraries: Linux Kernel & Device Drivers, Das U-Boot, Device Tree, Yocto Project.

Skills

- **Programming:** C, C++, Rust, Python, Bash, Matlab. (memory management, scheduling, interrupts).
- **On-device Machine Learning:** [Qualcomm AI Hub](#), [Qualcomm® AI Engine Direct](#), [Analog Devices AI](#), PyTorch, LiteRT, ONNX, etc.
- **Embedded & Real-Time:** FreeRTOS, CMSIS-DSP, CMSIS-NN, etc.
- **Systems & Linux:** Linux kernel & device drivers, board bring-up, bootloaders (U-Boot), Device Tree, cross-compilation, OS internals
- **eBPF:** CO-RE (BTF/libbpf), uprobes/kprobes, per-CPU maps & ring buffers, cilium/ebpf, bpf2go, bpftool.
- **Build, CI/CD & Packaging:** Yocto Project, RPM packaging, Jenkins, GitLab CI, GitHub Actions, git.
- **Containers:** Docker, Kubernetes.

Projects

eBPF-Based GPU Observability

Personal Project

Jun 2026 - Current

- Built a zero-instrumentation CUDA observability tool with eBPF, reporting device-memory leaks, kernel-launch statistics, and host/device transfer volume. ([GitHub](#))
- Engineered the hot path for high throughput/low latency.
- Wrote portable CO-RE eBPF loaded from a Go userspace via cilium/ebpf + bpf2go.

Programming Languages: C (eBPF), Go, CUDA C++.

Tools and Libraries: cilium/ebpf, libbpf/CO-RE, bpf2go, clang/LLVM, bpftool.

- Architected a real-time AI-assisted sonification algorithm for seizure detection in newborns and implemented it as a multi-threaded system on a low-power Arm Cortex-M4 based [AI microcontroller](#).
- Developed, quantized, and deployed neural networks on low-power accelerator.

Programming Languages: C, C++, Rust, Python, Matlab.

Tools and Libraries: CMSIS-DSP, CMSIS-NN, FreeRTOS, [Analog Devices AI](#).

Education

University College Cork

Cork, Ireland

[M.Sc - Electrical and Electronics Engineering](#)

May 2022 - May 2024

- Graduated with First Class Honours.
- Postgraduate research student in signal-processing, edge machine learning, and low-power embedded systems.
- Thesis: *Low-Power Real-Time Seizure Monitoring via AI-Assisted Sonification of Neonatal EEG*.

Hanoi University of Science and Technology

Hanoi, Vietnam

[Engineer - Control Engineering and Automation](#)

Oct 2015 - Aug 2020

- Minored in Instrumentation and Industrial Informatics.
- Graduated with GPA 3.37/4.0 and ranked 9th/155.
- ERASMUS+ Scholarship at Technical University of Munich, Munich, Germany, Winter Term 2019/20.

Publications

1. A real-time and ultra-low power implementation of an AI-assisted sonification algorithm for neonatal EEG
Tien Nguyen, Aengus Daly, Feargal O'Sullivan, Sergi Gomez Quintana, Andriy Temko, Emanuel Popovici
2023 9th International Workshop on Advances in Sensors and Interfaces (IWASI) 2023. doi:10.1109/IWASI58316.2023.10164463
2. Low-Power Real-Time Seizure Monitoring Using AI-Assisted Sonification of Neonatal EEG
Tien Nguyen, Aengus Daly, Sergi Gomez-Quintana, Feargal O'Sullivan, Andriy Temko, Emanuel Popovici
IEEE Transactions on Emerging Topics in Computing **13** 80–89. 2025. doi:10.1109/TETC.2024.3481035